Surprise-Taxonomy Pipeline and Agent Workflow

Explaining Markets competition entry · status as of 24 October 2026

Goal. Predict, for each earnings release, where the stock's next-day abnormal return (`car1`) will rank among all of the quarter's event reactions (a percentile in [0, 1]). The approach: have an LLM list every way the release differed from pre-release expectations, classify those surprises into a fixed set of dimensions, and turn them into features for a small trained model.

Status. Taxonomy formation (steps 1-3) is complete and the taxonomy is frozen at **v2**. The extraction and classification code for the remaining train and test events is written. The feature-and-model step and the live agent are planned but not built; the deployed `predict.py` still runs the earlier proposer/critic agent.

Part	Status	Where
Data split (train / test / discovery)	Done	src/suprise_dim/data.py, data/surprise/split.json
Free-form surprise extraction (500 discovery events)	Done	extraction.py, discovery_extractions.jsonl
Taxonomy proposal, assignment, review, refinement	Done, frozen at v2	grouping.py, data/surprise/taxonomy/
Reliability and stability checks	Done	grouping.py (reliability_check, agreement)
Extraction + classification for train_rest and test	Code written, run pending	dataset.py, notebook section 5
Combined one-row-per-surprise table	Code written	dataset.build_surprise_frame, surprises_v2.csv
Feature vector, trained model, live agent	Planned	not yet in code

1. Data and split

Source: `data/archive/EARNINGS_RELEASE_2026Q3.jsonl.gz`. An event is eligible when it has earnings-call facts, a non-empty earnings preview (written before the release), one focal asset and a realized return. Previews exist only in 2026Q3 (and 8 events in 2026Q4), so the pipeline uses 2026Q3 only.

Set	Events	How it is formed	Used for
Eligible 2026Q3	1,365	facts + preview + return + one focal asset	
Train	1,076	earliest 80% by release time; the cut falls on a release time so same-time events stay together	discovery and model fitting
Test	289	everything from 13 Aug 2026 21:00 UTC onward	one final out-of-time evaluation
Discovery	500	sampled from train, 50 per car1 decile (seed 42); the return is never shown to the LLM	taxonomy formation
train_rest	576	train events outside discovery	model fitting (planned)

2. Taxonomy formation pipeline

Steps run on the 500 discovery events, from `script/suprise_dim_discovery.ipynb` (sections 1-4). Every LLM step saves after each event or batch and resumes after an interruption.

1. Split *done*
Chronological 80/20 train/test on 2026Q3; 500 discovery events sampled from train, stratified by car1 decile.

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2. Free-form extraction (1 call per event) *done*
gpt-6-luna, medium effort, 8 workers. Input: numbered call facts + full preview. Output per surprise: description, direction, magnitude, fact evidence, expectation evidence, expectation source; plus items that came in as expected. 3,895 surprises, 7.8 per event.

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3a. Propose taxonomy v1 *done*
gpt-6.1-sol, medium. All items of 80 random events (about 610) plus 12 textbook seed dimensions. Rules: mutually exclusive, cross-industry, about what a surprise concerns (never its direction), 8-15 dimensions, no catch-all.

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3b. Assign every item *done*
gpt-6-luna, low effort, batches of 40, 8 workers. The answer format allows only the taxonomy's keys or 'other'. v1: 3,894 of 3,895 assigned, 1 in 'other'.

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3c. Review and reliability *done*
Read 'other', samples per dimension, signal per dimension. Reassign 400 random items in shuffled batches and compare: 90.7% same label, Cohen's kappa 0.896.

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3d. Refine to v2 and reassign *done*
gpt-6.1-sol revises from counts, 8 samples per dimension and 'other'. v2 keeps the 15 dimensions and tightens boundaries; v1 and v2 labels agree on 91.7% of items, the same as a plain rerun, so refinement has converged.

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3e. Stability *done*
An independent taxonomy from 80 other events (seed 1), 999 items assigned. NMI 0.64 overall; the gap is one design choice (it has no forward-guidance dimension). Without guidance items, 87.6% of items fall in matching families.

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Frozen: v2 (15 dimensions) *done*
Kept with forward_guidance as its own dimension.

Key results on the 500 discovery events

Score (Spearman with car1)	Value	95% range	Note
Net direction (positive minus negative items)	0.465	0.39 to 0.54	no taxonomy needed
Net weighted by magnitude (1/2/3)	0.492	0.42 to 0.56	beats plain net in 99% of resamples
Net of the first 3 items only	0.500	0.43 to 0.57	extraction order carries importance
Competition baselines (5 models)	0.357 to 0.426		net beats 4 of 5 clearly

Items are labelled 59.5% positive, 27.7% negative, 11.2% mixed and 1.6% unclear. The positive lean does not hurt rank correlation, but corporate-action dimensions (capital allocation, strategy, commercial) are labelled mostly positive and show no relation to returns.

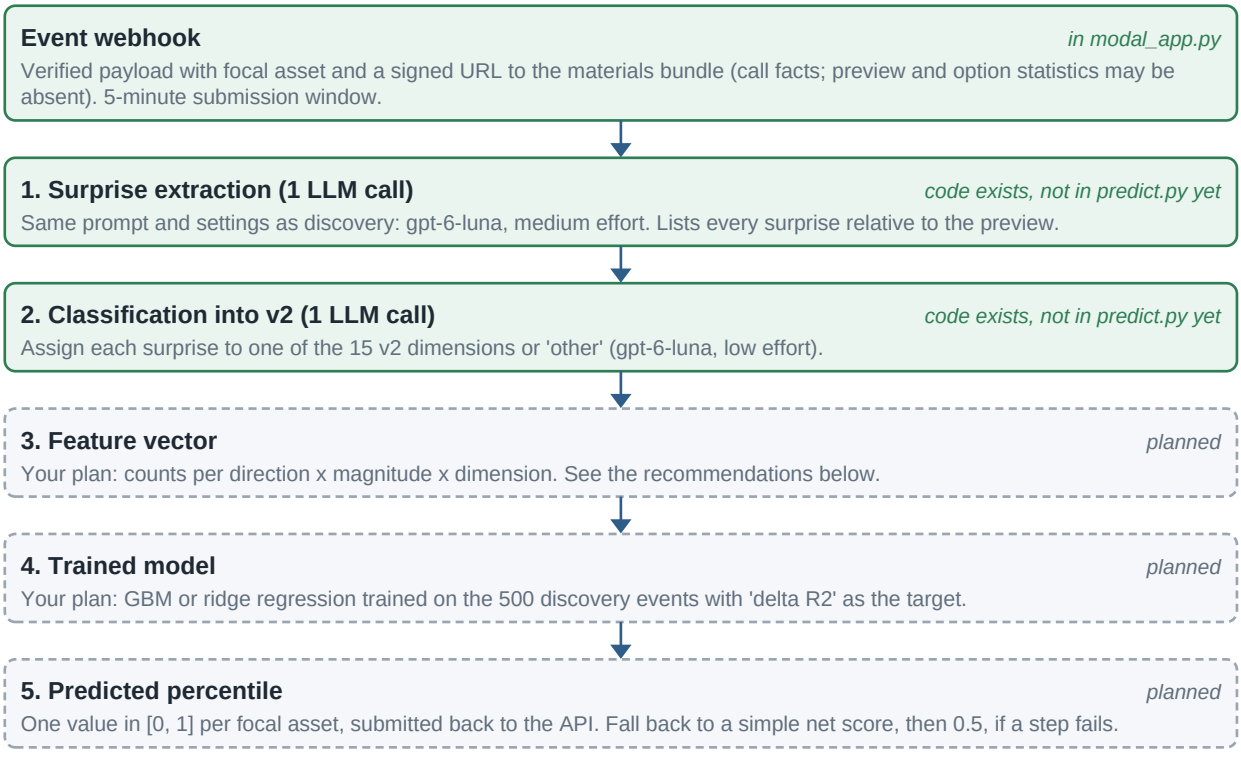
The frozen v2 taxonomy

Key	Dimension	Items	Share	Rho*
`forward_guidance`	Financial and operating outlook	685	17.6%	+0.340
`revenue`	Reported revenue and sales	605	15.5%	+0.352
`operating_kpis`	Recurring operating, demand and activity KPIs	523	13.4%	+0.185
`operating_profitability`	Operating profit, margins and returns	449	11.5%	+0.316
`bottom_line_earnings`	EPS and bottom-line earnings	269	6.9%	+0.241
`cash_balance_sheet_liquidity`	Cash flow, working capital, financial position	255	6.5%	+0.126
`commercial_economics`	Customer relationships, pricing, business mix	220	5.6%	+0.048
`development_execution_regulatory`	Development, execution, regulatory milestones	208	5.3%	-0.003
`capital_allocation`	Capital deployment and shareholder distributions	152	3.9%	+0.001
`costs_expenses`	Costs and expense efficiency	147	3.8%	+0.097
`strategy_corporate_events`	Corporate strategy, transactions, organization	122	3.1%	+0.015
`accounting_tax_special_items`	Accounting, reporting, taxes, special items	92	2.4%	+0.085
`external_environment`	Market environment, policy, market valuation	84	2.2%	+0.025
`financing_capital_structure`	Financing transactions and capital structure	67	1.7%	-0.035
`disclosure_delivery`	Delivery of specifically expected disclosures	16	0.4%	+0.041

* Spearman between each event's magnitude-weighted net score in that dimension and car1 (discovery events, exploratory only). One item is in 'other'. Reliability is weakest for the small dimensions (accounting 0.62, financing 0.67 on very few items); the large ones are 0.89 to 1.00.

3. Planned agent workflow

The agent that will replace the current proposer/critic graph in `predict.py`. Steps 1-2 reuse exactly the discovery pipeline (same model, prompt, effort and output format), so live features are produced the same way as training features.



Recommendations for steps 3-5

Topic	Issue	Recommendation
Feature vector	Direction x magnitude x dimension is 192 features; on discovery only 21 occur in 50+ events and 95% of entries are zero.	About 20-35 dense features: magnitude-weighted net per dimension (15), optionally positive and negative separately, plus total net, net of the first 3 items, item count, and the option-implied move.
Training data	500 events is small, and v2 was built from the same events, which inflates in-sample results.	Extract and classify the 576 train_rest events (code written), train on all 1,076 train events with time-ordered cross-validation, score test once.
Target	R2 is a measure of fit, not a per-event target.	Train on each event's within-quarter rank of car1 scaled to [0, 1]; evaluate with Spearman. If the competition also scores calibration, shrink toward 0.5.
Model	GBM on few rows overfits sparse features.	Ridge first. GBM only shallow, regularized and with monotone constraints, kept only if it beats ridge in cross-validation. Both must beat net (0.465) and weighted net (0.49).
Missing preview	build_user_message reads the preview directly; 42% of Q3 and 13 of 21 Q4 events have none, which would raise an error.	Add a no-preview prompt variant and train with a has-preview flag or a separate model.
Reproducibility	Features depend on the exact model and prompt.	Pin gpt-6-luna and the prompts once training starts; classification noise (kappa 0.90) favours dense per-dimension scores.

4. Settings and files

Step	Model	Effort	Batching / workers	Output
Extraction	gpt-6-luna	medium	1 event per call, 8 workers	data/surprise/<set>_extractions.jsonl
Taxonomy proposal / refinement	gpt-6.1-sol	medium	1 call per version	data/surprise/taxonomy/taxonomy_<version>.json
Assignment	gpt-6-luna	low	40 items per call, 8 workers	taxonomy/assignments_<version>[_<set>].jsonl
Combined table				data/surprise/surprises_v2.csv (one row per surprise)

Code: `src/suprise_dim/` (data, extraction, grouping, dataset, prompts, schemas). Driver: `script/suprise_dim_discovery.ipynb`.

5. Next steps

- Run notebook section 5: extract and classify train_rest (576) and test (289), then build `surprises_v2.csv`.
- Build features and fit ridge (and constrained GBM) on the 1,076 train events with time-ordered cross-validation.
- Freeze the model, then evaluate once on the 289 test events against the competition baselines; this also checks whether the model already knows outcomes.
- Add the no-preview path, then wire extraction, classification, features and model into `predict.py` with fallbacks.